

Methods of proof – 2

Induction: strengthening, dimension, descent

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Theoretical minimum.

Fact 1 (strengthen the statement). A statement is often easier to prove in a stronger form, because the induction hypothesis becomes stronger too. The standard example is an inequality with a constant: replace the constant by an explicit expression in n and the induction closes. If the step does not close, the correct reaction is to sharpen the statement, not to look for a longer computation.

Fact 2 (what to induct on). Besides the number of objects: the dimension of a space, the rank of a matrix, the degree of a polynomial, the number of prime factors, the size of a Jordan block, the number of vertices of a graph, the number of “bad” elements. Two parameters give a double induction, and a lexicographic order on pairs makes it a single one.

Theorem 1 (descent and the minimal counterexample). Every non-empty set of positive integers has a least element; hence a counterexample of least size exists and may be assumed to be minimal, and a construction producing a strictly smaller counterexample proves the statement. Over the reals this fails, and its place is taken by compactness (Methods of proof – 3).

Theorem 2 (transfinite induction and Zorn’s lemma). Every partially ordered set in which every chain has an upper bound has a maximal element. This is what produces a Hamel basis (Continuous functions – 6), a basis of an arbitrary vector space, a maximal ideal, and an extension of a linear functional; it is not needed when the object can be built by a countable procedure.

1. Prove by induction with a strengthened statement: $\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}$, and then $\prod_{k=1}^n \frac{2k-1}{2k} \leq \frac{1}{\sqrt{3n+1}}$; in both cases explain why the naive statement does not close and the sharpened one does.
2. Prove by induction on the dimension that every symmetric bilinear form has an orthogonal basis, and by induction on the rank that a matrix of rank r is a sum of r matrices of rank one (compare with Linear algebra – 2).
3. Prove Helly's theorem in $\mathbb{R}^n$ by induction on the number of sets, and explain which step uses the dimension and which uses the number of sets (compare with Convex – 1).
4. Prove by infinite descent that $\sqrt{2}$ is irrational, that $x^4 + y^4 = z^2$ has no positive integer solutions, and that a minimal counterexample argument proves the fundamental theorem of arithmetic.
5. Prove Zorn's lemma implies the existence of a basis in every vector space, and use transfinite induction to construct a set $A \subseteq \mathbb{R}^2$ meeting every line in exactly two points.